Aditya Sharma

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Computer Science graduate with professional experience in embedded C/C++ development, Python test automation, hardware communication protocols (CAN, RS-485, UDP), and full-stack application development.

Education

San Francisco State University — **Graduated: May 2025**
Bachelor of Science, Computer Science — *San Francisco, CA*

Relevant Coursework: Advanced Algorithms, Operating Systems, Computer Networks, Embedded Systems, Database Systems, App Development, Full-Stack Software Development

Experience

Junior Software Engineer — **Aug 2025 – Nov 2025**
Veethree Technologies — *Poole, United Kingdom*

- Designed and built an automated hardware test platform for the C4.3 marine/industrial display using Python (Flask) on a Raspberry Pi, replacing manual EMC test procedures and expanding hardware test coverage by 40%
- Implemented a multi-protocol communication stack (CAN, RS-485, Ethernet/UDP) to drive the device under test, parsing ACK/NAK response frames and rendering live results in a web UI
- Partnered with senior engineers to integrate the Python test tooling with embedded C firmware interfaces, enabling reliable automated communication across all three protocols
- Contributed embedded C/C++ features to the C3 display product using Veethree's internal SDK

Track Marshal — **May 2024 – Oct 2024**
K1 Speed — *Dublin, CA*

- Managed high-traffic live go-kart race operations with a focus on safety, incident response, and trackside efficiency
- Strengthened multitasking and team communication skills in a fast-paced, high-stakes environment

Projects

- **Peer Tutoring Platform (Capstone)** | *Node.js, Express, Handlebars, MySQL, AWS EC2* | GitHub — Designed and deployed a full-stack peer tutoring platform for SFSU students with bcrypt-secured session authentication, tutor search and scheduling, and file uploads; deployed to AWS EC2 via a GitHub Actions CI/CD pipeline
- **Embedded Autonomous Robot** | *C, PID control, GPIO/PWM* — Built a multithreaded autonomous car with PID-controlled line following and obstacle avoidance, including a PWM motor driver and a real-time IR line-detection sensor pipeline
- **Swift Ollama Chatbot** | *Swift, SwiftUI, Ollama* | GitHub — Built an iOS chatbot with selectable agents backed by locally running LLMs, eliminating external API calls to cut latency and keep user data on-device
- **SmugMug Slideshow** | *Python, REST, SQLite, JavaScript* | GitHub — Built a REST application that scrapes image URLs from a SmugMug sitemap and serves a randomized slideshow, with SQLite state persistence and a JS frontend
- **NBA Stats CLI** | *Java, JDBC, MySQL* — Command-line tool for querying and managing NBA statistics with dynamic filtering and reporting
- **Unix Shell** | *C, POSIX* — Implemented a Unix-like shell supporting command execution, pipes, I/O redirection, and built-ins
- **Java Tank Game** | *Java, Swing* — Two-player Tank Wars game with sprite animation, collision detection, and game-state management

Skills

Languages: Java, C/C++, Python, JavaScript, Swift, SQL, HTML/CSS
Frameworks & Libraries: Node.js, Express, Flask, Spring Boot, Hibernate, JUnit, REST APIs
Tools & Platforms: Git, GitHub Actions (CI/CD), AWS (EC2), Linux/Unix, Raspberry Pi, CAN/RS-485/UDP
Databases: MySQL, SQLite